

NES555 – Labs 1–3 Exam Cheat Sheet

Use this sheet for quick **study** and **PDF/print**. Section titles are large; commands are short.

Lab 1 – Packet Crafting & Network Analysis

Goal: Understand packets, flags, and reading captures.

Basic network (Kali)

- `ip a` → list interfaces + IPs.
- `ping -c 3 <IP>` → test target reachability.

Scapy basics

- Start: `scapy`
- List layers: `ls()`
- List commands: `lsc()`

Build packets

- IP header: `ip = IP(dst="IP")`
- TCP header: `tcp = TCP(dport=80)`
- Combine: `pkt = ip/tcp` → L3+L4 (OS fills MAC).
- Full frame: `pkt2 = Ether()/IP()/TCP()` → control L2 MAC.
- TTL auto: `del(ip.ttl)` → Scapy sets default TTL.

ICMP echo (ping)

- `sr1(IP(dst="IP")/ICMP()/"MSG")`
- ICMP Echo Reply if host is up, payload appears in data.

SYN probe (check port)

- `r = sr1(IP(dst="IP")/TCP(dport=80, flags="S"))`
- `SA` (SYN+ACK) → **open**.
- `R` (RST) → **closed**.
- No response → often **filtered/firewalled**.

Unusual flags (FIN+PSH)

- `sr1(IP(dst="IP")/TCP(dport=80, flags="FP"), timeout=2)`
- usually **no response** (packet dropped as out-of-state).

Capture & analysis

- Capture: `sudo tcpdump -i eth0 -s0 -w lab1.pcap`
- Wireshark filters: `icmp, tcp.port == 80, http`
- Right-click → **Follow** → **TCP Stream** for full convo.

Compare with nmap

- `nmap -sS -p80 IP` → expect `80/tcp open http` if Scapy SYN got `SA`.

Lab 2 – Passive Recon & Footprinting

Goal: Collect info **without active scanning**.

WHOIS (owner info)

- `whois domain` → registrar, creation/expiry, contacts.

DNS (technical records)

- `dig +noall +answer domain` → A records (IPs).
- `dig +short NS domain` → name servers.
- `dig +short MX domain` → mail servers.
- `dig +short -x IP` → PTR (reverse DNS).

Certificate Transparency (crt.sh)

- Browser: `https://crt.sh/?q=%25domain`
→ extra subdomains from TLS certs (e.g., `api`, `dev`, `staging`).

theHarvester (emails + hosts)

- `theHarvester -d domain -b bing -l 200 -f harvester_bing`
→ emails (`user@domain`) + hostnames from search engines.

Subdomains (multiple tools)

- `sublist3r -d domain -o s.txt`
- `amass enum -passive -d domain -o a.txt`
→ combine with crt.sh + theHarvester for bigger list.

recon-ng (OSINT framework)

- `recon-ng` → `workspaces create/select`
 - Install module, e.g.: `marketplace install recon/domains-hosts/bing_domain_web`
 - `modules load recon/domains-hosts/bing_domain_web`
 - `options set SOURCE domain` → `run` → `show hosts`
 - Export: `export csv ~/nes555/lab2/hosts.csv hosts`.
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Lab 3 – Scanning & Enumeration

Goal: Actively find live hosts, open ports, and services.

Capture scans

- `mkdir -p ~/nes555/lab3 && cd ~/nes555/lab3`
- `sudo tcpdump -i eth0 -w lab3.pcap &`

Host discovery

- `sudo nmap -sn NET/24 -oN ping_sweep.txt` → ping/ARP sweep.
- `sudo nmap -PR NET/24 -oN arp_sweep.txt` → ARP only (LAN, even if ICMP blocked).

TCP scans (Nmap)

- Quick common ports: `sudo nmap -Pn -sS -T4 -p1-1024 IP -oA quick`

- Full TCP + scripts: `sudo nmap -sS -sV -sC -p- -T4 IP -oA full`
- `-sS` SYN scan, `-sV` version, `-sC` default NSE scripts, `-p-` all ports.

UDP (selected ports)

- `sudo nmap -sU -p53,67,69,123,161 IP -oN udp.txt`.

hping3 (custom probes)

- ICMP: `sudo hping3 -1 -c4 IP`
 - SYN: `sudo hping3 -S -p80 -c3 IP`
 - ACK/FIN: `sudo hping3 -A -p80 -c3 IP`, `sudo hping3 -F -p80 -c3 IP`
- Interpret like Nmap: `SA` = open, `R` = closed, no reply = filtered.

Service checks (netcat + enum4linux)

- HTTP banner: ``echo -e "HEAD / HTTP/1.0`

" | nc IP 80 - SMB enum: `enum4linux -a IP > enum.txt` → users, shares, OS info.

Quick Concepts to Memorize

- **Passive recon (Lab 2):** no direct probing → WHOIS, DNS, CT logs, search engines, Shodan, recon-ng.
- **Active recon (Lab 3):** direct packets to target → Nmap, Masscan, hping3, netcat, enum4linux.
- **TCP flags:** S=SYN, A=ACK, F=FIN, P=PSH, R=RST.
- SYN → SYN+ACK (SA) = open; RST (R) = closed.
- No response → often filtered.
- **Scapy vs Nmap:** Scapy = manual packet crafting; Nmap = automated scanning + logic.